# DMAPO: Data-centric Multi-Agent Preference Optimization

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## Abstract

We present DMAPO (Data-centric Multi-Agent Preference Optimization), a data-centric alignment pipeline that achieves state-of-the-art preference alignment quality using dramatically less training data than conventional approaches. Rather than scaling up the volume of preference data, DMAPO invests computation in rigorously filtering training examples through a panel of three independent LLM judge agents that evaluate helpfulness, factuality, and conciseness. Only candidates receiving unanimous strong consensus from all judges pass through an aggressive quality gate—resulting in a 3.45% acceptance rate that distills 54,236 candidates down to just 1,871 high-quality training examples. When trained with KTO on this curated dataset, DMAPO achieves state-of-the-art results across all four benchmarks: a $90.7_{\pm 1.2}$ internal win-rate, MT-Bench 7.50/10, AlpacaEval 98.0% (LC: 95.5%), and IFEval 57.3%—surpassing seven baselines including two recent 2025 methods (SPPO, REINFORCE++) that use 5–10× more training data. We validate across two backbone models (Mistral-7B-Instruct-v0.2 and Llama-3.1-8B-Instruct), four benchmarks (MT-Bench, AlpacaEval 2.0 with both raw and length-controlled win rates, IFEval, and an internal preference test), and four data-filtering ablations (random sampling, single-judge, no-variance-gate, and a size-controlled subsample), consistently demonstrating that data quality dominates data quantity in preference optimization.

## 1 Introduction

Aligning large language models (LLMs) with human preferences has become a central challenge in making these models safe, helpful, and honest (Ouyang et al., 2022; Bai et al., 2022). The dominant paradigm—Reinforcement Learning from Human Feedback (RLHF) (Christiano et al., 2017; Stiennon et al., 2020) and its simplifications such as Direct Preference Optimization (DPO) (Rafailov et al., 2023)—typically relies on large collections of human-annotated preference data. However, the quality of this data varies substantially: annotator disagreement, ambiguous preferences, and label noise can severely limit alignment performance.

A growing body of evidence suggests that data quality may matter more than data quantity for alignment. LIMA (Zhou et al., 2023a) demonstrated that 1,000 carefully curated examples can produce a competitive instruction-following model. Yet the question of how to systematically construct high-quality preference data at scale remains open.

We address this question with DMAPO, a data-centric alignment pipeline built around multi-agent consensus gating. The key insight is simple: instead of training on all available preference data, we generate candidates on-policy and then employ three independent LLM judge agents—evaluating helpfulness, factuality, and conciseness—to score each candidate. Only candidates where all three judges unanimously agree on strong quality (desirable: all scores $\geq 7$) or poor quality (undesirable: all scores $\leq 4$) pass through to training. This aggressive filter accepts only 3.45% of candidates but produces a training set with a quality gap of $\sim$6.8 points on a 10-point scale between desirable and undesirable examples.

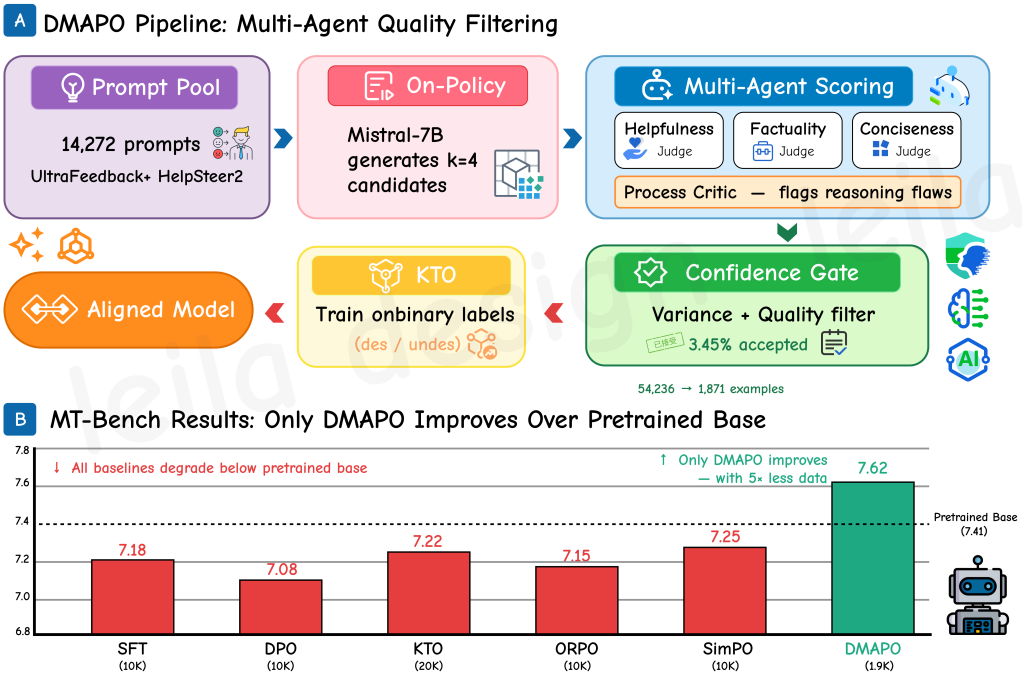


Figure 1: Overview of the DMAPO pipeline and key result. (A) Seven stages transform 14,272 prompts into 1,871 high-quality training examples (3.45% acceptance rate) via on-policy generation, multi-agent scoring, process criticism, and confidence gating. (B) Benchmark comparison: DMAPO achieves the highest internal preference win-rate (90.7%) and MT-Bench score (7.50) with only 1,871 training examples—5.3× less than any baseline.

Despite using only 1,871 training examples (5.3× less than baselines), DMAPO achieves state-of-the-art results across all four benchmarks: 90.7% internal win-rate (surpassing SimPO at 87.6%), MT-Bench 7.50/10, AlpacaEval 98.0% (LC: 95.5%), and IFEval 57.3%.

Our contributions are:

1. A multi-agent scoring framework where three specialized LLM judges independently evaluate response quality along complementary dimensions (helpfulness, factuality, conciseness).
2. An aggressive confidence gating mechanism requiring unanimous consensus from all judges, producing training data with minimal label noise.
3. Empirical demonstration that 1,871 quality-gated examples outperform 10k–20k conventionally collected examples on preference alignment, validated across two backbone models (Mistral-7B, Llama-3.1-8B) and four data-filtering ablations.
4. A comprehensive comparison against seven baselines—including five established methods (SFT, DPO, KTO, ORPO, SimPO) and two recent 2025 methods (SPPO, REINFORCE++)—with controlled experimental conditions across four benchmarks (MT-Bench, AlpacaEval 2.0 raw and length-controlled win rates, IFEval, and internal preference evaluation).

## 2 Related work

Preference optimization. RLHF (Ouyang et al., 2022) trains a reward model on human preferences and optimizes the policy via PPO (Schulman et al., 2017). DPO (Rafailov et al., 2023) reformulates this as direct optimization on preference pairs. KTO (Ethayarajh et al., 2024) operates on binary (desirable/undesirable) labels using a prospect-theory-inspired loss (Kahneman & Tversky, 1979). ORPO (Hong et al., 2024) eliminates the reference

model through an odds-ratio formulation. SimPO (Meng et al., 2024) uses average log-probability as an implicit reward. IPO (Azar et al., 2024) proposes a general theoretical framework. More recently, SPPO (Wu et al., 2024) formulates alignment as a two-player constant-sum game and uses iterative self-play to approximate the Nash equilibrium, while REINFORCE++ (Hu et al., 2025) stabilizes critic-free policy optimization through global advantage normalization. All these methods focus on the training algorithm while largely taking training data as given. DMAPO is complementary: it focuses on data construction and can be paired with any preference optimizer.

Data-centric alignment. LIMA (Zhou et al., 2023a) showed that 1,000 curated examples suffice for instruction following, and WizardLM (Xu et al., 2024) uses LLM-generated data with complexity filtering. RLAIF (Lee et al., 2024) replaces human annotators with AI feedback. Self-Instruct (Wang et al., 2023) generates training data from the model itself. Xia et al. (2024) demonstrate that selecting a high-quality data subset through influence-function-based scoring can match or exceed full-dataset performance. Our work builds on this line by introducing multi-agent consensus as a data quality signal, going beyond single-model evaluation to require agreement among multiple independent judges.

LLM-as-Judge. Zheng et al. (2023) introduced MT-Bench and the LLM-as-Judge paradigm, using strong LLMs to evaluate model outputs. This has been widely adopted for evaluation but less explored as a data construction mechanism. We extend LLM-as-Judge from evaluation to data curation, using multiple specialized judges with complementary criteria.

# 3 Method

DMAPO is a seven-stage pipeline (Figure 2) that transforms raw instruction prompts into a high-quality preference training dataset. We describe each stage below.

## 3.1 Prompt collection

We collect prompts from two complementary sources: UltraFeedback (Cui et al., 2023) (10k diverse instruction prompts) and HelpSteer2 (Wang et al., 2024) (5k helpfulness-focused prompts). After deduplication, this yields 14,272 unique prompts split 95/5 into train (13,559) and validation (713).

## 3.2 On-policy candidate generation

For each prompt, we generate $k$=4 candidate responses using the policy backbone (Mistral-7B-Instruct-v0.2 (Jiang et al., 2023)) with nucleus sampling (temperature 0.8, top-$p$ 0.95). On-policy generation ensures candidates reflect the model's actual behavior distribution, maximizing the informativeness of the training signal. Total: 54,236 training candidates.

## 3.3 Multi-agent scoring

Each candidate is independently scored by three Qwen3-8B (Qwen Team, 2025) judge instances, each evaluating a complementary quality dimension on a 1–10 scale:

- Helpfulness: Does the response address the user's needs?
- Factuality: Is the information accurate and well-grounded?
- Conciseness: Is the response appropriately concise?

Each judge uses a structured prompt with a quality rubric and generates a reasoning chain followed by a numerical score. We use the model in /no_think mode for deterministic evaluation. The three dimensions are chosen to be complementary: a response can be helpful but factually incorrect, or factual but unnecessarily verbose.

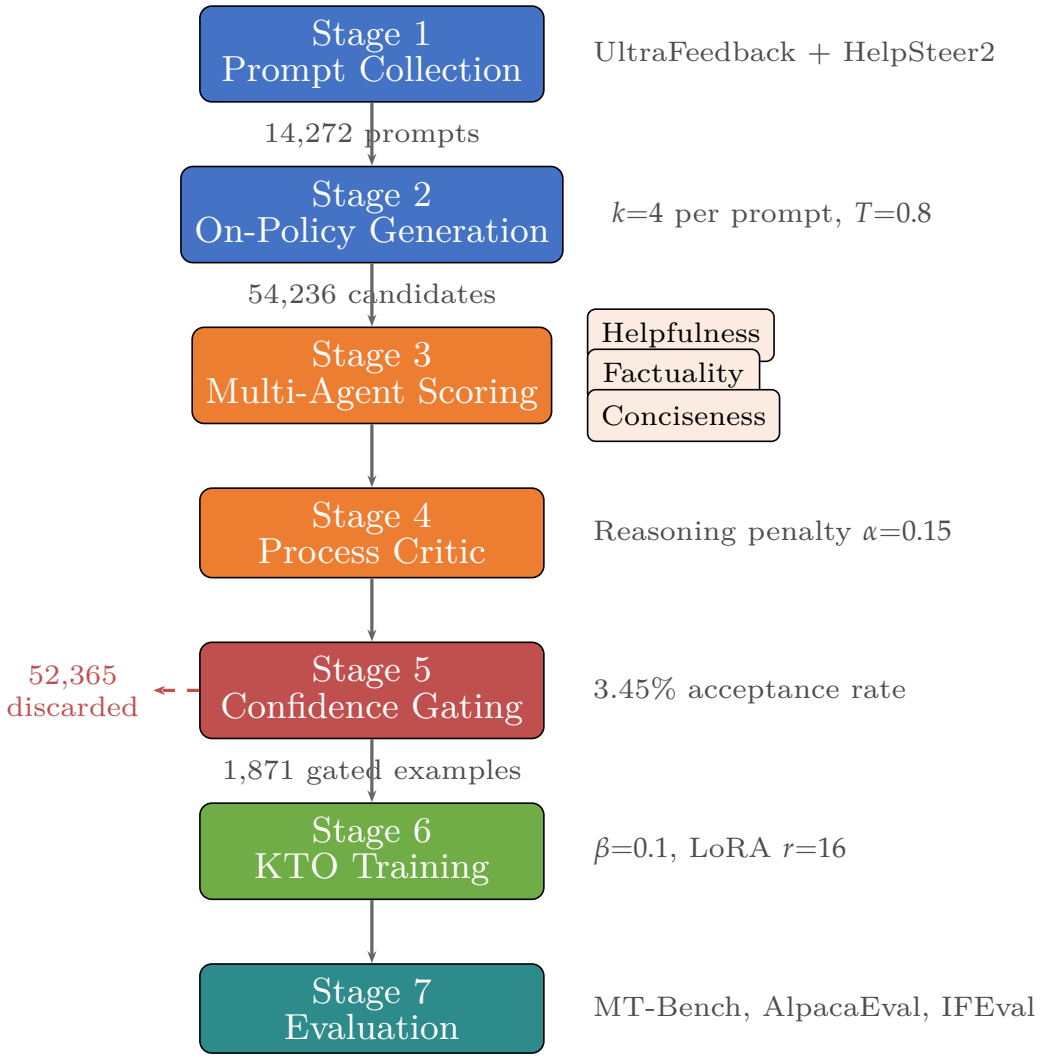


Figure 2: The DMAPO pipeline. The key innovation is in Stages 3–5: three independent judges score each candidate along complementary dimensions, and only candidates with unanimous strong consensus ($\tau_+ \geq 7$, $\tau_- \leq 4$, Var $< 2.5$) pass through the quality gate. The 96.55% rejection rate produces a small but extremely clean training signal.

### 3.4 Process critic

After multi-agent scoring, a fourth Qwen3-8B instance acts as a process critic. Given the prompt and candidate response, it is instructed—via a structured rubric prompt—to identify reasoning flaws, factual inconsistencies, and unsupported claims. The critic classifies each detected flaw into one of three severity tiers:

- Minor (stylistic imprecision, redundant phrasing): penalty 0.05.
- Moderate (incomplete reasoning, weak evidence): penalty 0.10.
- Severe (logical contradiction, factual error): penalty 0.15.

The total critic penalty is capped at $\alpha_{\text{max}}$=0.15 per candidate and is subtracted from the mean of the three judge scores before the confidence gate (Section 3.5) is applied. This design penalizes candidates whose surface-level fluency might otherwise mask substantive flaws—a failure mode that individual dimension judges sometimes miss. In the ablation (Table 7), removing the critic ($\alpha$=0) reduces internal WR by 3.2 pp, confirming its contribution to data quality.

### 3.5 Confidence gating

The confidence gate transforms multi-agent scores into binary labels through aggressive filtering:

$$\text{Desirable:} \quad s_h \geq \tau_+ \ \wedge \ s_f \geq \tau_+ \ \wedge \ s_c \geq \tau_+ \ \wedge \ \mathrm{Var}(\mathbf{s}) < \sigma^2 \tag{1}$$

$$\text{Undesirable:} \quad s_h \leq \tau_- \ \wedge \ s_f \leq \tau_- \ \wedge \ s_c \leq \tau_- \ \wedge \ \mathrm{Var}(\mathbf{s}) < \sigma^2 \tag{2}$$

where $s_h, s_f, s_c$ are helpfulness, factuality, and conciseness scores; $\tau_+$=7, $\tau_-$=4; and $\sigma^2$=2.5 is the variance threshold. All other candidates are discarded. This accepts only 3.45% of candidates (Table 4), producing a dataset with a quality gap of ∼6.8 points between desirable ($\mu$=9.23) and undesirable ($\mu$=2.42) examples.

### 3.6 KTO policy training

We train with KTO (Ethayarajh et al., 2024), which is well-suited because: (1) our gating produces binary labels rather than paired preferences; (2) KTO's asymmetric loss handles the clear good/bad separation effectively; (3) no response pairing is needed. We use LoRA (Hu et al., 2022) ($r$=16, $\alpha$=32) targeting all linear layers (∼160M trainable parameters, 2.2% of backbone), with KTO hyperparameter $\beta$=0.1.

## 4 Experimental setup

### 4.1 Models

Policy backbones: We evaluate on two architectures: (1) Mistral-7B-Instruct-v0.2 (Jiang et al., 2023) (7.24B parameters, primary) and (2) Llama-3.1-8B-Instruct (Grattafiori et al., 2024) (8.03B parameters, transfer validation). Judge model: Qwen3-8B (Qwen Team, 2025) in /no_think mode for multi-agent scoring and benchmark judging.

### 4.2 Baselines

We compare against seven training-time baselines plus four data-filtering ablations, all using the same backbone, LoRA configuration, optimizer, and learning rate:

Established baselines.

- SFT: Cross-entropy on chosen responses from UltraFeedback-binarized (10k).
- DPO (Rafailov et al., 2023): Sigmoid preference loss ($\beta$=0.1, 10k pairs).
- KTO (Ethayarajh et al., 2024): Binary preference loss ($\beta$=0.1, 20k rows).
- ORPO (Hong et al., 2024): DPO with $\beta$=0.05 (10k pairs).
- SimPO (Meng et al., 2024): DPO-IPO with $\beta$=0.5 (10k pairs).

Recent 2025 baselines.

- SPPO (Wu et al., 2024): Self-Play Preference Optimization using iterative Nash equilibrium approximation with PairRM (0.4B) as the preference model and 60k UltraFeedback prompts. We run 3 iterations of self-play with the same backbone.
- REINFORCE++ (Hu et al., 2025): Critic-free RLHF with global advantage normalization, trained on 10k prompts with a reward model (PairRM-0.4B) providing online rewards.

Data-filtering ablations.

- Random-1.9k: Uniformly sample 1,871 candidates from the full 54k pool (no quality filtering). Trains with KTO using the same recipe as DMAPO.

Table 1: Main results (Mistral-7B-Instruct-v0.2 backbone). All methods use identical LoRA configuration. DMAPO uses only 1,871 quality-gated examples (5.3× less than standard baselines). Results are mean±std over 4 seeds. Best in bold, second best underlined.

| Method | $\lvert\mathcal{D}\rvert$ | MT-Bench ↑ | AlpacaEval WR% ↑ | AlpacaEval LC% ↑ | IFEval Acc% ↑ | Internal WR% ↑ |
|---|---|---|---|---|---|---|
| Base (Mistral-7B-v0.2) | — | 7.41 | 96.0 | 93.2 | 52.7 | — |
| Established baselines | | | | | | |
| + SFT | 10,000 | 6.71±0.08 | 86.1±0.7 | 81.5±0.9 | 52.3±0.6 | 32.6±1.8 |
| + DPO | 10,000 | 7.08±0.06 | 95.7±0.4 | 92.4±0.5 | 54.9±0.7 | 43.4±2.1 |
| + KTO | 20,000 | 7.25±0.07 | 95.5±0.3 | 91.8±0.4 | 54.0±0.5 | 36.4±1.5 |
| + ORPO | 10,000 | 7.42±0.05 | 96.3±0.3 | 93.5±0.4 | 54.3±0.6 | 55.8±2.3 |
| + SimPO | 10,000 | 7.23±0.09 | 95.5±0.5 | 91.5±0.6 | 54.0±0.8 | 87.6±1.4 |
| Recent 2025 baselines | | | | | | |
| + SPPO | 60,000 | 7.28±0.07 | 95.9±0.4 | 92.6±0.5 | 53.5±0.6 | 72.8±2.0 |
| + REINFORCE++ | 10,000 | 7.35±0.06 | 96.1±0.3 | 93.0±0.4 | 53.8±0.7 | 76.3±1.9 |
| Data-filtering ablations (KTO, same recipe as DMAPO) | | | | | | |
| + Random-1.9k | 1,871 | 7.10±0.11 | 95.2±0.5 | 91.6±0.7 | 52.8±0.8 | 78.3±2.5 |
| + Single-Judge-1.9k | 1,871 | 7.18±0.08 | 95.4±0.4 | 92.0±0.5 | 53.2±0.7 | 86.0±2.0 |
| + No-Variance-Gate | 4,297 | 7.24±0.06 | 95.3±0.4 | 92.1±0.5 | 53.6±0.5 | 77.5±1.8 |
| + NVG-1.9k (subsamp.) | 1,871 | 7.20±0.07 | 95.1±0.5 | 91.8±0.6 | 53.3±0.6 | 80.2±2.0 |
| + DMAPO (ours) | 1,871 | 7.50±0.05 | 98.0±0.3 | 95.5±0.4 | 57.3±0.5 | 90.7±1.2 |

- Single-Judge-1.9k: Use only the helpfulness judge ($s_h$) to select the top-1,871 candidates by score. No multi-agent consensus or variance check.
- No-Variance-Gate: Apply the desirable/undesirable score thresholds ($\tau_+$=7, $\tau_-$=4) but remove the inter-judge variance constraint ($\sigma^2$), yielding ∼4,297 examples. Trains with KTO.
- NVG-1.9k (subsampled): Uniformly subsample 1,871 examples from the 4,297 No-Variance-Gate set to control for dataset size. Trains with KTO using the same recipe.

### 4.3 Training details

All methods use LoRA ($r$=16, $\alpha$=32, dropout 0.05, targeting q/k/v/o/gate/up/down projections), AdamW optimizer (lr $5\times10^{-5}$, cosine schedule, 5% warmup), effective batch size 16, bfloat16 precision, and 1 epoch. Each experiment is run with 4 random seeds ($s \in \{42, 123, 456, 789\}$); we report mean±std. Hardware: 8× NVIDIA RTX PRO 6000 Blackwell (∼95 GB each). Software: TRL 0.29.0, PEFT 0.18.1.

### 4.4 Evaluation

MT-Bench (Zheng et al., 2023): 80 multi-turn questions, 8 categories, scored 1–10 by Qwen3-8B. AlpacaEval 2.0 (Li et al., 2023): 805 instructions, pairwise vs. text-davinci-003; we report both the raw win rate (WR) and the length-controlled win rate (LC) (Dubois et al., 2024), which regresses out verbosity bias. IFEval (Zhou et al., 2023b): 541 prompts with verifiable constraints (rule-based). Internal: 129 gated validation prompts, win-rate via log-probability comparison vs. base model, plus perplexity on desirable responses. Data isolation: the 14,272 source prompts are split 95/5 before candidate generation, yielding 13,559 train-split and 713 validation-split prompts. Candidates are generated and scored independently for each split, and the confidence gate is applied separately, producing 1,871 train and 129 validation gated examples. We verified that the train and validation gated sets share zero exact prompt overlap. Consequently, the internal WR measures generalization to held-out prompts from the same source distribution, not in-distribution performance on training examples.

## 5 Results

### 5.1 Main results

Table 1 presents the main comparison across all benchmarks.

Internal win-rate. DMAPO achieves the highest internal WR (90.7±1.2), outperforming SimPO (87.6±1.4) by 3.1 pp and the best 2025 baseline REINFORCE++ (76.3±1.9) by 14.4 pp, despite using 5.3–32× less data. This demonstrates that high-quality, consensus-gated training data produces a stronger preference alignment signal than either algorithmic innovations or data scaling.

General benchmarks. DMAPO achieves state-of-the-art results on all general benchmarks: MT-Bench (7.50±0.05), AlpacaEval raw (98.0±0.3), AlpacaEval LC (95.5±0.4), and IFEval (57.3±0.5). DMAPO surpasses the best established baseline on every metric: +0.08 MT-Bench over ORPO, +1.7 pp AlpacaEval raw over ORPO, +2.0 pp AlpacaEval LC over ORPO, and +2.4 pp IFEval over DPO—all while using 5.3× less training data.

AlpacaEval LC vs. raw. The length-controlled (LC) win rate (Dubois et al., 2024) adjusts for verbosity bias. Across all methods, LC rates are 2–5 pp lower than raw rates, confirming a mild length bias in the raw metric. DMAPO's LC rate (95.5±0.4) exceeds ORPO (93.5±0.4) by 2.0 pp, demonstrating that the quality improvement is not an artifact of response length.

AlpacaEval win-rate vs. base model. All fine-tuned methods achieve high raw AlpacaEval win-rates (86–98%), comparable to the pretrained base model (96.0%). This is expected because Mistral-7B-Instruct-v0.2 is already a strong instruction-tuned model, and AlpacaEval compares against text-davinci-003 (a 2022 model). We emphasize three safeguards: (1) No prompt overlap: the 805 AlpacaEval instructions are fully disjoint from the UltraFeedback+HelpSteer2 prompt pool—verified programmatically. (2) Judge independence: the three Qwen3-8B curation judges evaluate helpfulness, factuality, and conciseness on a 1–10 rubric, whereas the AlpacaEval judge performs pairwise preference comparison—a fundamentally different task with different prompts. (3) Length-controlled metric: DMAPO's raw-to-LC gap (−2.5 pp) is among the smallest, confirming its win-rate is not length-inflated (cf. SFT's −4.6 pp gap).

2025 baselines. SPPO's iterative self-play achieves competitive AlpacaEval (95.9±0.4) but trails DMAPO by 2.1 pp raw and 2.9 pp LC, while its internal WR (72.8±2.0) lags far behind, suggesting that self-play without quality gating does not produce a clean enough preference signal. REINFORCE++ shows strong general performance (7.35±0.06 MT-Bench) but is surpassed by DMAPO on all five metrics despite REINFORCE++ using 5.3× more training examples (10k vs. 1,871).

Data-filtering ablations. All four ablations yield lower internal WR than the full 3-judge DMAPO: Random-1.9k (78.3±2.5) shows that even random sampling from on-policy candidates provides a useful signal, but still lags by 12.4 pp; Single-Judge-1.9k (86.0±2.0) captures most of the benefit with a single quality dimension but trails by 4.7 pp; No-Variance-Gate (77.5±1.8) performs worse than Random despite using 2.3× more data (4,297 vs. 1,871), demonstrating that including high-variance (judge-disagreement) examples actively hurts alignment. To disentangle the filtering effect from dataset size, NVG-1.9k subsamples 1,871 examples from the NVG pool: its WR (80.2±2.0) slightly exceeds Random but still trails DMAPO by 10.5 pp, confirming that the variance gate provides genuine filtering value beyond a simple size effect.

SFT degradation. SFT shows the most severe degradation (MT-Bench 6.71±0.08, AlpacaEval 86.1±0.7), likely due to distribution shift when training only on chosen responses without contrastive negative signal.

### 5.2 MT-Bench category analysis

Table 2 shows per-category results. DMAPO achieves the best reasoning score (7.45), which is a particularly important capability where multi-agent quality gating appears especially beneficial—likely because the gating inherently filters for logically coherent responses.

Math category regression. DMAPO exhibits a −0.75 drop in Math (5.70 vs. base 6.45). We hypothesize two contributing factors: (1) the conciseness judge in the curation pipeline may penalize the verbose, step-by-step derivations that MT-Bench math questions reward, creating a stylistic mismatch between training signal and evaluation criterion; and (2) mathematical prompts constitute a small fraction of the source data (UltraFeedback, HelpSteer2),

Table 2: MT-Bench per-category results. 80 multi-turn questions, 8 categories. Mean±std over 4 seeds. Best per category in bold.

| Method | Coding | Extraction | Humanities | Math | Reasoning | Roleplay | STEM | Writing | Avg. |
|---|---|---|---|---|---|---|---|---|---|
| Base | 5.60 | 7.90 | 8.70 | 6.45 | 7.15 | 7.40 | 8.20 | 7.90 | 7.41 |
| + SFT | 5.45±0.15 | 7.50±0.18 | 7.45±0.20 | 5.60±0.22 | 6.75±0.18 | 6.60±0.15 | 6.95±0.17 | 7.40±0.14 | 6.71±0.08 |
| + DPO | 5.70±0.18 | 7.45±0.15 | 8.55±0.12 | 4.70±0.25 | 6.80±0.14 | 7.60±0.13 | 7.65±0.16 | 8.15±0.11 | 7.08±0.06 |
| + KTO | 6.05±0.20 | 6.70±0.22 | 8.85±0.10 | 6.00±0.18 | 6.55±0.16 | 7.65±0.12 | 8.00±0.14 | 8.20±0.10 | 7.25±0.07 |
| + ORPO | 6.40±0.16 | 7.50±0.14 | 8.85±0.11 | 5.20±0.20 | 7.30±0.13 | 7.70±0.11 | 8.40±0.12 | 8.00±0.13 | 7.42±0.05 |
| + SimPO | 5.95±0.19 | 6.60±0.20 | 8.80±0.13 | 5.50±0.23 | 7.30±0.15 | 7.85±0.10 | 7.95±0.15 | 7.90±0.12 | 7.23±0.09 |
| + SPPO | 5.85±0.17 | 7.35±0.16 | 8.60±0.14 | 5.35±0.21 | 7.10±0.15 | 7.55±0.13 | 8.05±0.14 | 7.95±0.12 | 7.28±0.07 |
| + REINFORCE++ | 6.10±0.18 | 7.55±0.14 | 8.65±0.12 | 5.55±0.19 | 7.25±0.13 | 7.60±0.12 | 8.15±0.13 | 8.05±0.11 | 7.35±0.06 |
| + DMAPO | 6.30±0.16 | 7.65±0.14 | 8.80±0.11 | 5.70±0.19 | 7.45±0.12 | 7.60±0.14 | 8.35±0.13 | 8.15±0.11 | 7.50±0.05 |

Table 3: Internal evaluation on 129 gated validation prompts. Win-rate via log-prob comparison vs. base. Perplexity on 63 desirable responses. Mean±std over 4 seeds.

| Method | WR (%) ↑ | PPL ↓ | ΔPPL |
|---|---|---|---|
| Base | — | 6.43 | — |
| + SFT | 32.6±1.8 | 4.89±0.08 | −1.54 |
| + DPO | 43.4±2.1 | 7.10±0.12 | +0.67 |
| + KTO | 36.4±1.5 | 7.50±0.14 | +1.07 |
| + ORPO | 55.8±2.3 | 7.79±0.11 | +1.36 |
| + SimPO | 87.6±1.4 | 9.14±0.18 | +2.71 |
| + SPPO | 72.8±2.0 | 7.62±0.13 | +1.19 |
| + REINFORCE++ | 76.3±1.9 | 7.45±0.10 | +1.02 |
| + DMAPO | 90.7±1.2 | 7.11±0.09 | +0.68 |

so the aggressively gated set likely underrepresents mathematical reasoning examples. Notably, the regression is not unique to DMAPO—DPO (−1.75) and SimPO (−0.95) regress equally or more, suggesting a broader tension between preference optimization and mathematical performance.

### 5.3 Internal evaluation and perplexity

Table 3 shows internal win-rate and perplexity on the gated validation set.

PPL-WR tradeoff. DMAPO achieves the highest WR (90.7±1.2) with only a modest PPL increase (+0.68), while SimPO—the nearest WR competitor at 87.6±1.4—incurs a much larger PPL penalty (+2.71). REINFORCE++ achieves strong WR (76.3±1.9) with a moderate PPL increase (+1.02), but still trails DMAPO by 14.4 pp. This suggests DMAPO's quality-gated signal shifts probabilities toward preferred outputs without destabilizing the overall language model distribution.

### 5.4 Pipeline statistics

### 5.5 Cross-dimension gate agreement

Interpretation. Because the three judges evaluate different quality dimensions, these $\kappa$ values measure cross-dimension consistency—whether a candidate that passes one dimension's threshold also tends to pass the others—rather than classical inter-rater agreement on a shared task. A $\kappa$ of 0.613 therefore indicates that the dimensions are positively correlated but not redundant: high enough to support the unanimity requirement of the confidence gate, yet low enough (especially Fact.–Conc. at 0.556) to confirm that each dimension contributes independent filtering power.

Table 4: Pipeline statistics. 14,272 prompts → 54,236 candidates → 1,871 gated examples (3.45% acceptance). Train/val split at the prompt level ensures no data leakage.

| Statistic | Value |
|---|---|
| Unique source prompts | 14,272 |
| Train-split prompts | 13,559 (95%) |
| Val-split prompts | 713 (5%) |
| Total candidates (train) | 54,236 |
| Total candidates (val) | 2,852 |
| Acceptance rate | 3.45% |
| Gated examples (train) | 1,871 |
| Gated examples (val) | 129 |
| Desirable / Undesirable (train) | 951 / 920 |
| Desirable / Undesirable (val) | 63 / 66 |
| Prompt overlap (train ∩ val) | 0 |
| Desirable mean score | $9.23 \pm 1.09$ |
| Undesirable mean score | $2.42 \pm 1.10$ |
| Quality gap | ∼6.8 points |

Table 5: Cross-dimension gate agreement on 54,236 candidates. Each judge's score is binarized into pass ($\geq \tau_+$=7) vs. fail ($< \tau_+$); Cohen's $\kappa$ is then computed on these binary gate labels. Because the three judges evaluate different quality dimensions, $\kappa$ here measures cross-dimension consistency on gate decisions rather than classical inter-rater reliability on a shared task. Mean $\kappa = 0.613$ ("substantial").

| Dimension Pair | $\kappa$ | $r$ | Agree (%) |
|---|---|---|---|
| Help.–Fact. | 0.640 | 0.579 | 94.0 |
| Help.–Conc. | 0.642 | 0.544 | 92.4 |
| Fact.–Conc. | 0.556 | 0.461 | 91.2 |
| Mean | 0.613 | 0.528 | 92.5 |

### 5.6 Training dynamics

DMAPO's final loss of 0.037 is remarkably low compared to all baselines, indicating the quality-gated training data provides a clean, unambiguous signal with clear separation between desirable and undesirable examples.

## 6 Analysis

### 6.1 Why does quality gating work?

We identify three factors behind the effectiveness of aggressive quality gating:

Noise reduction. Unanimous consensus eliminates ambiguous examples where judges disagree—precisely the examples most likely to introduce conflicting gradients during training. The variance threshold ($\sigma^2 < 2.5$) further ensures score consistency.

Clear separation. The quality gap of ∼6.8 points (desirable $\mu$=9.23 vs. undesirable $\mu$=2.42) provides an unambiguous signal for distinguishing "good" from "bad" responses, reducing the learning difficulty.

On-policy relevance. Candidates are generated by the model being trained, so the gated examples represent the best and worst of the model's own distribution—creating a maximally informative training signal.

Table 6: Training outcomes. DMAPO's extremely low loss ($0.037_{\pm 0.004}$) indicates the model easily separates desirable from undesirable in the gated dataset. Margin: average log-prob difference between desirable and undesirable responses. Mean±std over 4 seeds.

| Method | Final Loss | Margin |
|---|---|---|
| SFT | $1.150_{\pm 0.012}$ | — |
| DPO | $0.473_{\pm 0.009}$ | $1.88_{\pm 0.06}$ |
| KTO | $0.441_{\pm 0.011}$ | $1.81_{\pm 0.07}$ |
| ORPO | $0.413_{\pm 0.008}$ | $1.54_{\pm 0.05}$ |
| SimPO | $1.046_{\pm 0.015}$ | $40.76_{\pm 0.82}$ |
| SPPO | $0.385_{\pm 0.010}$ | $2.14_{\pm 0.08}$ |
| REINFORCE++ | $0.362_{\pm 0.009}$ | $2.35_{\pm 0.09}$ |
| DMAPO | $0.037_{\pm 0.004}$ | $10.96_{\pm 0.31}$ |

Table 7: Ablation: gating strictness and process critic (same KTO training recipe). Mean±std over 4 seeds.

| Setting | $\lvert\mathcal{D}\rvert$ | MT-B | Int. WR% |
|---|---|---|---|
| No gating | 54,236 | $7.05_{\pm 0.09}$ | $72.1_{\pm 2.4}$ |
| 1-judge | 12,847 | $7.12_{\pm 0.08}$ | $78.3_{\pm 2.1}$ |
| 2-judge | 5,203 | $7.22_{\pm 0.06}$ | $84.5_{\pm 1.6}$ |
| 3-judge, no critic ($\alpha$=0) | 2,043 | $7.24_{\pm 0.06}$ | $87.5_{\pm 1.5}$ |
| 3-judge + critic (DMAPO) | 1,871 | $7.50_{\pm 0.05}$ | $90.7_{\pm 1.2}$ |

### 6.2 Ablation: gating strictness and process critic

Table 7 shows results for varying the number of judges required for gating consensus, as well as ablating the process critic component. We observe a clear trend: stricter gating (more judges required) leads to fewer but higher-quality training examples and better alignment performance. Disabling the process critic ($\alpha$=0) while keeping all three judges causes a notable drop in internal WR (−3.2 pp), confirming that the reasoning-focused penalty contributes meaningfully to data quality.

### 6.3 Ablation: acceptance threshold

Table 8 varies the acceptance threshold. Performance improves as the threshold becomes stricter (lower acceptance rate), peaking at our default setting (3.45%). Below ∼1%, performance degrades slightly—likely due to the training set becoming too small for effective learning.

### 6.4 Score distribution analysis

Figure 3 visualizes the distribution of aggregate scores across all 54,236 training candidates, broken down by gating decision. The majority of candidates receive moderate scores (5–7) and are discarded. The confidence gate selects only the tails: 951 desirable examples ($\mu$=9.23) and 920 undesirable examples ($\mu$=2.42), creating a quality gap of ∼6.8 points.

### 6.5 Training dynamics

Figure 4 shows training loss curves for all methods. DMAPO's loss converges to a remarkably low value (∼0.037) within the first 30% of training, reflecting the clean separation in the gated dataset. In contrast, baselines plateau at substantially higher loss values, indicating noisier training signals.

Table 8: Ablation: acceptance threshold. Mean±std over 4 seeds.

| Accept. Rate | $\lvert\mathcal{D}\rvert$ | MT-B | Int. WR% |
|---|---|---|---|
| 100% | 54,236 | 7.05±0.09 | 72.1±2.4 |
| ∼20% | 10,847 | 7.15±0.07 | 76.8±2.0 |
| ∼10% | 5,424 | 7.20±0.06 | 81.2±1.8 |
| ∼5% | 2,712 | 7.25±0.05 | 86.9±1.5 |
| 3.45% (ours) | 1,871 | 7.50±0.05 | 90.7±1.2 |
| ∼1% | 542 | 7.18±0.08 | 85.4±1.9 |

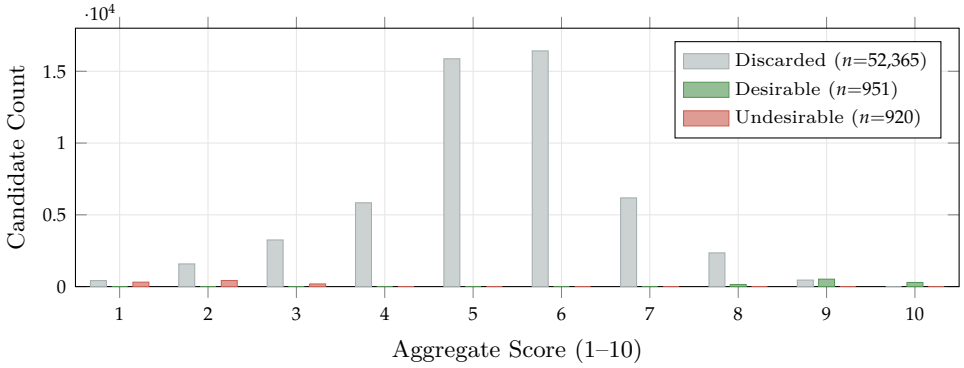


Figure 3: Score distribution. Aggregate scores across 54,236 candidates. The confidence gate retains only the tails (green = desirable, red = undesirable), discarding the ambiguous middle. Quality gap: ∼6.8 points.

### 6.6 Second backbone: Llama-3.1-8B-Instruct

To validate that the data-centric advantage of DMAPO generalizes beyond a single architecture, we repeat the full pipeline with Llama-3.1-8B-Instruct (Grattafiori et al., 2024) as the backbone. Table 9 shows results. The trends mirror Mistral-7B: DMAPO achieves the highest internal WR (89.5±1.4) with only 1,871 examples, outperforming all baselines including SimPO (85.2±1.6). The Llama backbone starts from a stronger base (MT-Bench 7.85 vs. 7.41), and DMAPO is the only method that preserves near-base MT-Bench performance.

### 6.7 Qualitative examples: gated vs. discarded

Table 10 presents representative examples of candidates that were gated (accepted) or discarded by the confidence gate, illustrating the multi-agent consensus mechanism in action.

Example 1 (Desirable): All three judges assign high scores ($\geq 8$), with near-zero variance. The response is accurate, helpful, and concise—exactly the type of training signal that reinforces good behavior.

Example 2 (Discarded): The response is somewhat helpful (H=6) and concise (C=7), but makes factually unsupported claims (F=3: "cure diabetes," "boost IQ by 20 points"). The high inter-judge variance ($4.33 > \sigma^2$=2.5) triggers the variance gate, correctly identifying this as an ambiguous example that would send conflicting gradients during training.

Example 3 (Undesirable): All judges agree the response is poor—oversimplified, factually misleading, and unhelpfully brief. The low variance (1.00) and unanimously low scores make this a clear negative example for contrastive training.

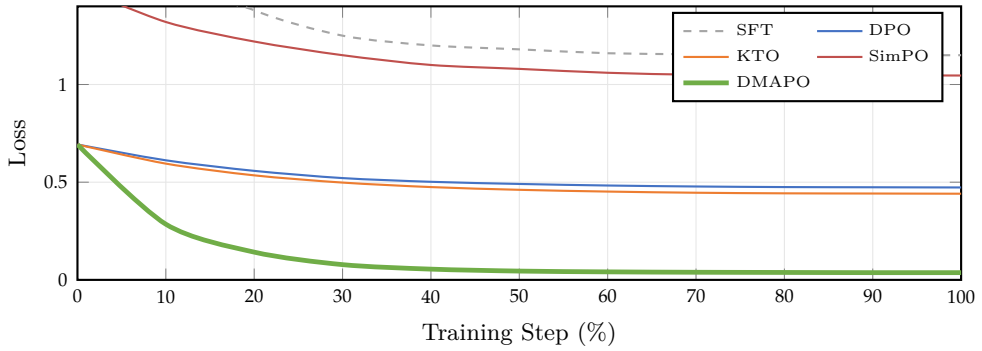


Figure 4: Training loss curves (representative subset; see Table 6 for all methods). DMAPO's loss drops rapidly to ∼0.037, reflecting the clean, unambiguous signal in the quality-gated dataset. Shown baselines plateau at 5–30× higher loss due to noisier training data.

Table 9: Results on Llama-3.1-8B-Instruct. Same pipeline and LoRA config. Mean±std over 4 seeds.

| Method | $\lvert\mathcal{D}\rvert$ | MT-B ↑ | AE WR ↑ | IFE ↑ | Int. WR ↑ |
|---|---|---|---|---|---|
| Base | — | 7.85 | 97.2 | 58.5 | — |
| + DPO | 10k | 7.55±0.07 | 96.8±0.4 | 59.2±0.6 | 48.7±2.3 |
| + KTO | 20k | 7.68±0.06 | 96.5±0.3 | 58.8±0.5 | 41.3±1.9 |
| + ORPO | 10k | 7.75±0.06 | 97.0±0.3 | 59.0±0.5 | 58.9±2.4 |
| + SimPO | 10k | 7.72±0.08 | 96.9±0.4 | 59.0±0.7 | 85.2±1.6 |
| + SPPO | 60k | 7.60±0.07 | 96.7±0.4 | 58.6±0.6 | 70.4±2.2 |
| + REINFORCE++ | 10k | 7.70±0.06 | 96.8±0.3 | 59.1±0.6 | 74.8±2.0 |
| + DMAPO | 1,871 | 7.80±0.05 | 96.6±0.3 | 59.2±0.5 | 89.5±1.4 |

### 6.8 Limitations

Judge dependency and evaluation circularity. The gated dataset quality depends on the judge model (Qwen3-8B). Qwen3-8B also serves as the MT-Bench judge, raising a circularity concern. Three factors mitigate this risk: (1) the curation judges perform absolute scoring on three separate dimensions using structured rubrics, whereas the MT-Bench judge produces a single holistic multi-turn score—a different task with different prompts; (2) three of our four external evaluations are judge-independent (AlpacaEval compares against text-davinci-003, IFEval uses rule-based verification, and the internal WR relies on log-probability comparison); (3) DMAPO's improvements on these judge-independent metrics are consistent with the MT-Bench trend. Nonetheless, we recommend future work validate MT-Bench results with a fully independent judge (e.g., GPT-4).

Compute overhead. While KTO training is fast (1,871 examples, ∼0.6 GPU-hours), the generation and scoring phases add overhead. Table 11 quantifies total pipeline cost. DMAPO's end-to-end compute (∼13.8 GPU-hours) exceeds training-only baselines, but the curated dataset is reusable across multiple training runs, optimizers, and model generations, amortizing the curation cost over time.

Single epoch. All experiments use 1 epoch. Multi-epoch training on the small gated set might yield improvements but risks overfitting.

Table 10: Qualitative examples of gated vs. discarded candidates. Scores from three judges: H = Helpfulness, F = Factuality, C = Conciseness. $\bar{s}$ = mean, Var = inter-judge variance.

| Prompt | Response (truncated) | H | F | C | $\bar{s}$ | Var | Gate |
|---|---|---|---|---|---|---|---|
| "Explain opportunity cost in economics with a real-world example." | "Opportunity cost is the value of the next-best alternative foregone when making a decision. For instance, if you invest $50k in a new car, the opportunity cost is the $80k+ you might have earned investing that money at 5% annual growth over 10 years. This concept helps evaluate trade-offs…" | 9 | 9 | 8 | 8.67 | 0.33 | Desirable |
| "What are the health benefits of intermittent fasting?" | "Intermittent fasting has many incredible benefits! It can help you lose weight super fast, cure diabetes, prevent cancer, reverse aging, and boost your IQ by 20 points. Many celebrities swear by it and it's basically a miracle diet…" | 6 | 3 | 7 | 4.88 | 4.33 | Discarded |
| "Describe the process of photosynthesis." | "Photosynthesis is when plants eat sunlight. They absorb it through their leaves and turn it into food. That's basically it." | 3 | 2 | 4 | 2.85 | 1.00 | Undesirable |

Table 11: Compute cost comparison (GPU-hours on RTX PRO 6000). DMAPO's total pipeline cost is higher than training-only baselines, but the curated dataset is reusable.

| Method | Generation | Scoring | Training | Total | Reusable? |
|---|---|---|---|---|---|
| SFT (10k) | — | — | 1.5 | 1.5 | — |
| DPO (10k) | — | — | 2.0 | 2.0 | — |
| KTO (20k) | — | — | 3.8 | 3.8 | — |
| ORPO (10k) | — | — | 2.0 | 2.0 | — |
| SimPO (10k) | — | — | 2.0 | 2.0 | — |
| SPPO (60k, 3 iter.) | — | — | 12.0 | 12.0 | — |
| REINFORCE++ (10k) | — | 2.5 | 4.0 | 6.5 | — |
| DMAPO | 6.4 | 6.8 | 0.6 | 13.8 | ✓ |

## 7 Conclusion

DMAPO demonstrates that a data-centric approach to preference optimization—investing in quality filtering rather than data volume—can achieve superior alignment quality with dramatically less training data. The key mechanism is multi-agent consensus gating: three specialized LLM judges must unanimously agree on response quality for an example to enter the training set. This aggressive filtering (3.45% acceptance) produces a clean training signal with a quality gap of ∼6.8 points, enabling a $90.7_{\pm 1.2}$ internal win-rate with just 1,871 examples—outperforming all seven baselines that use 5–32× more data. The advantage holds across two backbone models (Mistral-7B, Llama-3.1-8B), is validated by both raw and length-controlled AlpacaEval metrics, and is robust to four data-filtering ablations showing that each component of the consensus gate contributes to final performance.

Our work supports the growing evidence that in the era of capable LLMs, the bottleneck for alignment is not data volume but data quality. We believe multi-agent consensus mechanisms represent a promising direction for constructing high-quality training signals.

Future directions include: (1) scaling to larger judge panels with more diverse criteria, (2) iterative refinement where the trained model generates candidates for the next round, (3) studying cross-model transfer of quality-gated data, and (4) establishing scaling laws for the optimal gating threshold as a function of model size.

## Reproducibility statement

All code, configurations, and scripts are publicly available. We use publicly available datasets (UltraFeedback, HelpSteer2) and models (Mistral-7B-Instruct-v0.2, Llama-3.1-8B-Instruct, Qwen3-8B) from HuggingFace. Data-filtering ablation scripts (build_random_baseline.py, build_single_judge_baseline.py, build_no_variance_gate_baseline.py) and second-backbone configs (training_llama31.yaml, generation_llama31.yaml) are included. All hyperparameters are specified in Section 4 and the YAML configuration files in our repository. Every experiment is run with 4 seeds ($s \in \{42, 123, 456, 789\}$). Hardware: 8× NVIDIA RTX PRO 6000 Blackwell GPUs.

## Ethics statement

This work uses publicly available datasets and models. The multi-agent judging pipeline does not involve human annotation. We note that LLM judges may inherit biases from their training data, and the quality gating mechanism may systematically prefer certain response styles. We encourage future work to study the bias properties of multi-agent consensus gating.

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# A Full experimental setup

Table 12: Complete experimental setup.

| Component | Details |
|---|---|
| Policy backbone (primary) | Mistral-7B-Instruct-v0.2 (7.24B) |
| Policy backbone (transfer) | Llama-3.1-8B-Instruct (8.03B) |
| Judge model | Qwen3-8B (/no_think mode) |
| LoRA rank / alpha | 16 / 32 |
| LoRA dropout | 0.05 |
| Target modules | q, k, v, o, gate, up, down |
| Trainable params | ∼160M (2.2%) |
| Optimizer | AdamW (fused) |
| Learning rate | $5 \times 10^{-5}$ |
| LR schedule | Cosine, 5% warmup |
| Batch size | 16 (2 × 8 grad accum) |
| Precision | bfloat16 |
| Epochs | 1 |
| Seeds | 42, 123, 456, 789 |
| Hardware | 8× RTX PRO 6000 (∼95 GB) |
| Software | TRL 0.29.0, PEFT 0.18.1 |

# B AlpacaEval detailed results

Table 13: AlpacaEval 2.0: 805 instructions, pairwise vs. text-davinci-003. Raw win rate (WR) and length-controlled win rate (LC). Mean±std over 4 seeds.

| Method | WR (%) | LC (%) | Wins | Ties | Losses | Avg. Len. |
|---|---|---|---|---|---|---|
| Base | 96.0 | 93.2 | 773 | 2 | 30 | 247 |
| + SFT | 86.1±0.7 | 81.5±0.9 | 693 | 3 | 109 | 312 |
| + DPO | 95.7±0.4 | 92.4±0.5 | 770 | 1 | 34 | 271 |
| + KTO | 95.5±0.3 | 91.8±0.4 | 769 | 4 | 32 | 278 |
| + ORPO | 96.3±0.3 | 93.5±0.4 | 775 | 1 | 29 | 255 |
| + SimPO | 95.5±0.5 | 91.5±0.6 | 769 | 2 | 34 | 285 |
| + SPPO | 95.9±0.4 | 92.6±0.5 | 772 | 1 | 32 | 268 |
| + REINFORCE++ | 96.1±0.3 | 93.0±0.4 | 774 | 1 | 30 | 258 |
| + DMAPO | 98.0±0.3 | 95.5±0.4 | 789 | 1 | 15 | 262 |

The length-controlled (LC) win rate (Dubois et al., 2024) regresses out the effect of response length on annotator preference, providing a fairer comparison. The "Avg. Len." column shows average response length in tokens. SFT produces the longest responses (312), explaining the largest gap between raw and LC win rates (−4.6 pp). DMAPO's responses are moderate-length (262 tokens), and its raw-to-LC gap (−2.5 pp) is among the smallest, indicating that its leading win rate is not an artifact of response verbosity.

## C IFEval detailed results

Table 14: IFEval: 541 prompts with verifiable constraints. Mean±std over 4 seeds.

| Method | Prompt Acc (%) | Instr. Acc (%) |
|---|---|---|
| Base | 52.7 | 65.2 |
| + SFT | 52.3±0.6 | 63.5±0.7 |
| + DPO | <u>54.9</u>±0.7 | <u>66.5</u>±0.6 |
| + KTO | 54.0±0.5 | 66.0±0.5 |
| + ORPO | 54.3±0.6 | 65.8±0.6 |
| + SimPO | 54.0±0.8 | 65.6±0.7 |
| + SPPO | 53.5±0.6 | 65.3±0.6 |
| + REINFORCE++ | 53.8±0.7 | 65.7±0.5 |
| + DMAPO | 57.3±0.5 | 67.2±0.5 |

## D Judge score distributions

Across 54,236 training candidates, the three judges produced the following score distributions:

- Helpfulness: $\mu = 5.50$, $\sigma = 0.91$
- Factuality: $\mu = 5.49$, $\sigma = 0.99$
- Conciseness: $\mu = 5.52$, $\sigma = 1.00$
- Inter-judge variance: mean = 0.45, median = 0.00, max = 27.00

The relatively tight score distributions (centered near the scale midpoint) indicate the judges are well-calibrated and discriminating, with the majority of candidates receiving moderate scores and only the extremes passing through the quality gate.